

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & \left\| \left\{ \sum_{k \in \mathbb{Z}} \Re^{-k\beta} |D_k(a)| \right\} \right\|_{L_{\omega}^{p'}} \\ & \leq \left\| \left\{ \sum_{k=k_0+1}^{\infty} \Re^{-k\beta} |D_k(a)| \right\} \right\|_{L_{\omega}^{p'}} + \left\| \left\{ \sum_{k=-\infty}^{k_0} \Re^{-k\beta} |D_k(a)| \right\} \right\|_{L_{\omega}^{p'}} \\ & =: I_1 + I_2. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$W_{\vec{k}, \vec{l}}(\vec{y}, \vec{x}) = \overline{W_{\vec{k}, \vec{l}}(\vec{x}, \vec{y})}, \tilde{W}_{\vec{k}, \vec{l}}(\vec{y}, \vec{x}) = \overline{\tilde{W}_{\vec{k}, \vec{l}}(\vec{x}, \vec{y})}, \check{W}_{\vec{k}, \vec{l}}(\vec{y}, \vec{x}) = \overline{\check{W}_{\vec{k}, \vec{l}}(\vec{x}, \vec{y})},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{cases} 0 &= \varphi'(t) + [A + \bar{A} + \beta(C + \bar{C})]^\top \varphi(t) \\ & - \Pi_5^\top(t) \Pi_2^{-1}(t) \Pi_4^\top(t), \\ \varphi(T) &= 0 \end{cases}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$z \frac{d}{dz} + \begin{pmatrix} b_1(z) & b_2(z) & b_3(z) & \dots & b_{n-1}(z) & b_n(z) \\ 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \end{pmatrix} \leftrightarrow \frac{d^n}{d\tau^n} + \bar{b}_1(\tau) \left(\frac{d^{n-1}}{d\tau^{n-1}} \right) + \dots + \bar{b}_n(\tau),$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \nabla E &= \nabla P + \frac{1}{2}((u-c)^2 + v^2) \nabla \varrho + \varrho(-cu_x + uu_x + vv_x, -cu_y + uu_y + vv_y) \\ &+ \varrho \mathbf{g} + gy \nabla \varrho \\ &= ((\sqrt{\varrho}(u-c))_y - (\sqrt{\varrho}v)_x) (-\sqrt{\varrho}v, \sqrt{\varrho}(u-c)) + gy \nabla \varrho, \end{aligned}$$